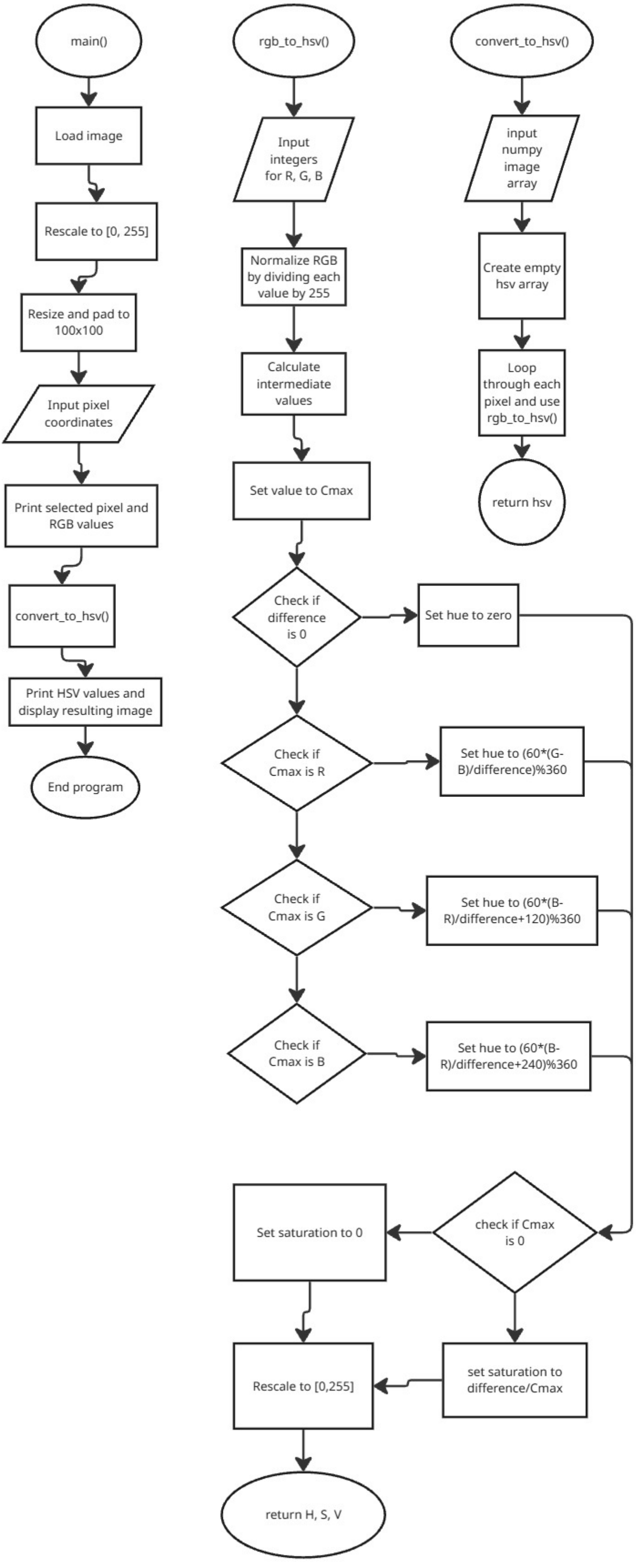

main()
Load image
Rescale to [0, 255]
Resize and pad to 100x100
Input pixel coordinates
Print selected pixel and RGB values
convert_to_hsv()
Print HSV values and display resulting image
End program
rgb_to_hsv()
Input integers for R, G, B
Normalize RGB by dividing each value by 255
Calculate intermediate values
Set value to Cmax
Check if difference is 0
Set hue to zero
Check if Cmax is R
Set hue to (60*(G-B)/difference)%360
Check if Cmax is G
Set hue to (60*(B-R)/difference+120)%360
Check if Cmax is B
Set hue to (60*(B-R)/difference+240)%360
check if Cmax is 0
Set saturation to 0
set saturation to difference/Cmax
Rescale to [0,255]
return H, S, V
convert_to_hsv()
input numpy image array
Create empty hsv array
Loop through each pixel and use rgb_to_hsv()
return hsv